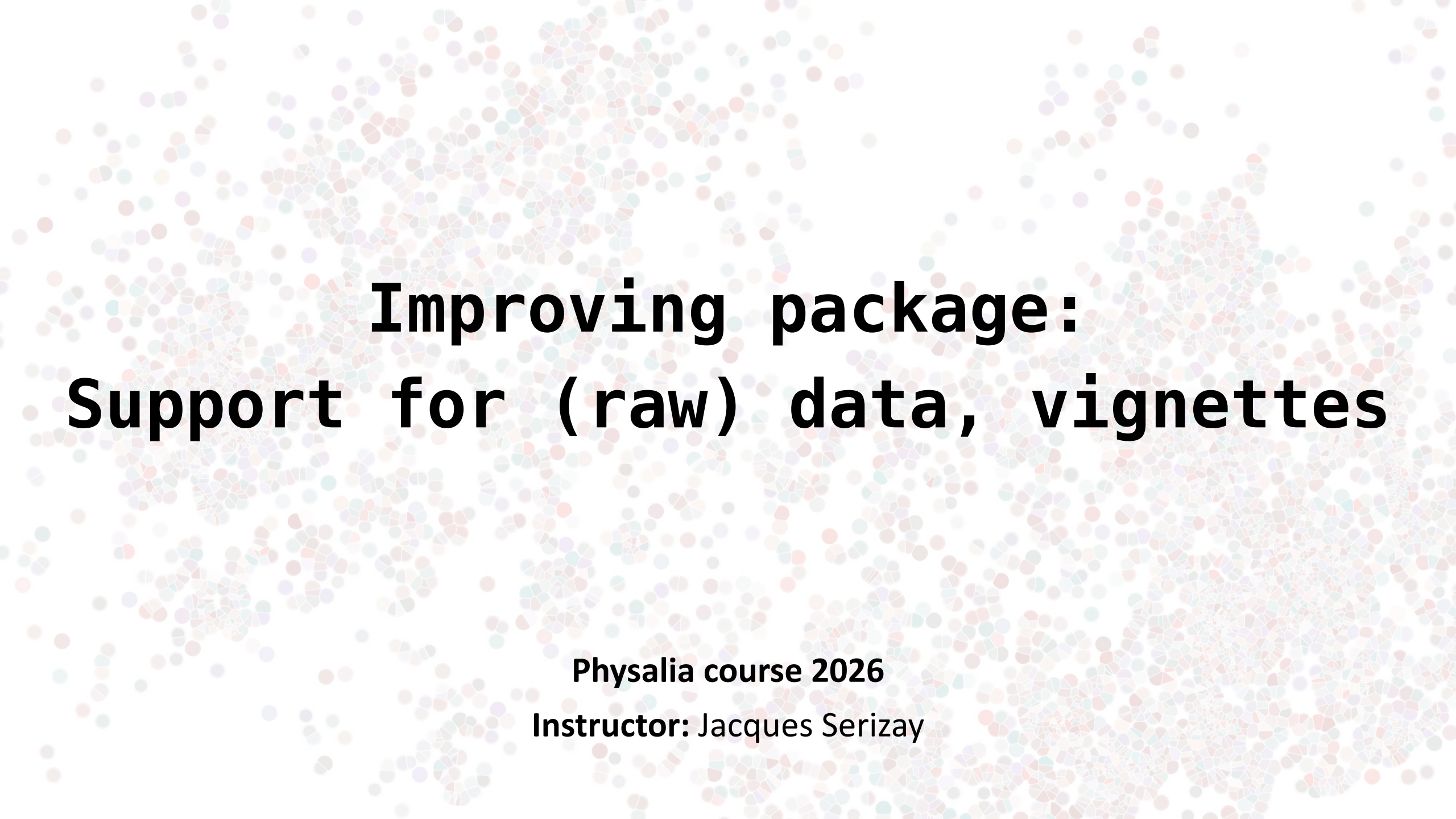


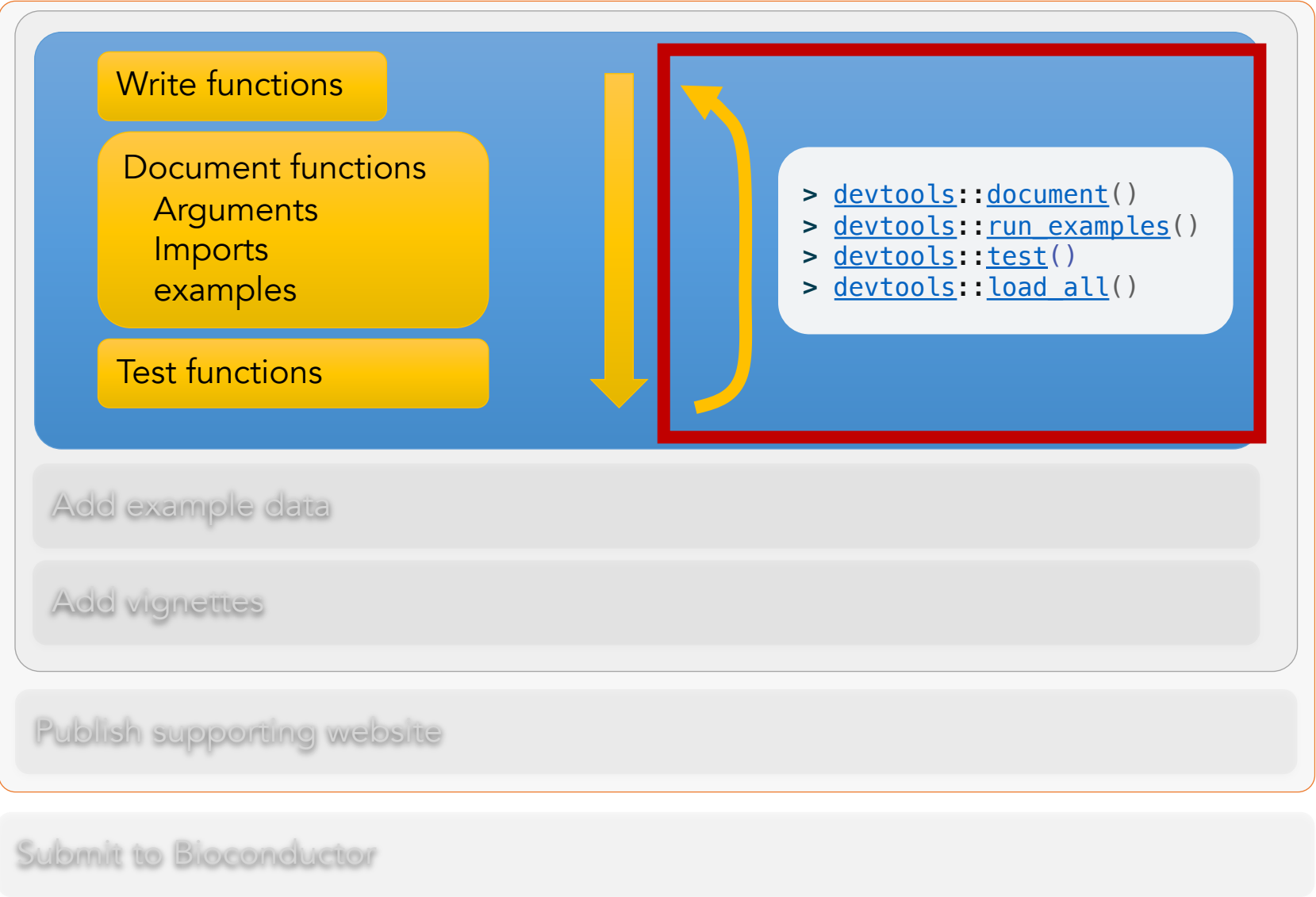
The background of the slide is a dense, abstract pattern of small, semi-transparent circles and triangles in various colors including red, blue, green, yellow, and purple. These shapes are scattered across the entire white background, creating a textured, mosaic-like effect.

Improving package: Support for (raw) data, vignettes

Physalia course 2026

Instructor: Jacques Serizay

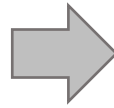
Standard package content



```
myPackage/  
  R/  
    functions.R  
    utils.R  
  man/  
    myfunction.Rd  
  tests/  
    testthat.R  
    testthat/  
      test-myfun.R  
DESCRIPTION  
README.md  
NAMESPACE  
NEWS  
LICENSE
```

Standard package content

```
myPackage/  
  R/  
    functions.R  
    utils.R  
  man/  
    myfunction.Rd  
  tests/  
    testthat.R  
    testthat/  
      test-myfun.R  
DESCRIPTION  
README.md  
NAMESPACE  
NEWS  
LICENSE
```



```
myPackage/  
  R/  
    functions.R  
    utils.R  
  man/  
    myfunction.Rd  
  tests/  
    testthat.R  
    testthat/  
      test-myfun.R  
  inst/  
    extdata/  
      <raw-data-file>  
  data/  
    <data>.Rda  
  vignettes/  
    myPackage.Rmd  
DESCRIPTION  
NAMESPACE  
README.md  
NEWS  
LICENSE
```

Why providing data in your package?

Data shipped with your package are meant to:

1. Provide a means to run examples and demonstrate package functionalities in vignettes;
2. Directly enable analysis (in “data” packages)

Adding data to package

2 types of "data":

- Raw: e.g. genomic files (bed, bigwig, bam, ...) or other (tables, text files, ...)
- Processed: `.Rda` files, containing R objects to be loaded in memory in R.

Package size limits

Watch out! Your package should be < 5Mb. Genomic files (particularly) can expand in size very quickly.

Be cautious!

Package size limits

Watch out! Your package should be < 5Mb. Genomic files (particularly) can expand in size very quickly.
Be cautious!

This is a rather strict requirement for Bioconductor. Under only very few specific circumstances will the BioC core members allow you to exceed this (e.g. developing specific packages coordinated by the BioC team itself).

Package size limits

Watch out! Your package should be < 5Mb. Genomic files (particularly) can expand in size very quickly.
Be cautious!

This is a rather strict requirement for Bioconductor. Under only very few specific circumstances will the BioC core members allow you to exceed this (e.g. developing specific packages coordinated by the BioC team itself).

Unrelated but still worth mentioning:

Don't forget, git never forgets! If you add a dataset and commit/push it to your git repo, it will stay there forever. Even after deleting it, it will still be in your `.git` local folder and in git memory (because you should be able to recover it back, since everything is reversible in git). This usually results in enormous `.git` folders... Watch out for storage space!

Raw data

Raw data can be virtually any file, but it has to be relevant for the package development.

The main reason to include such files is when a key part of a package's functionality is to act on an external file (e.g. ``readr``, ``vroom``, ...).

Raw data is stored in ``inst/extdata``.

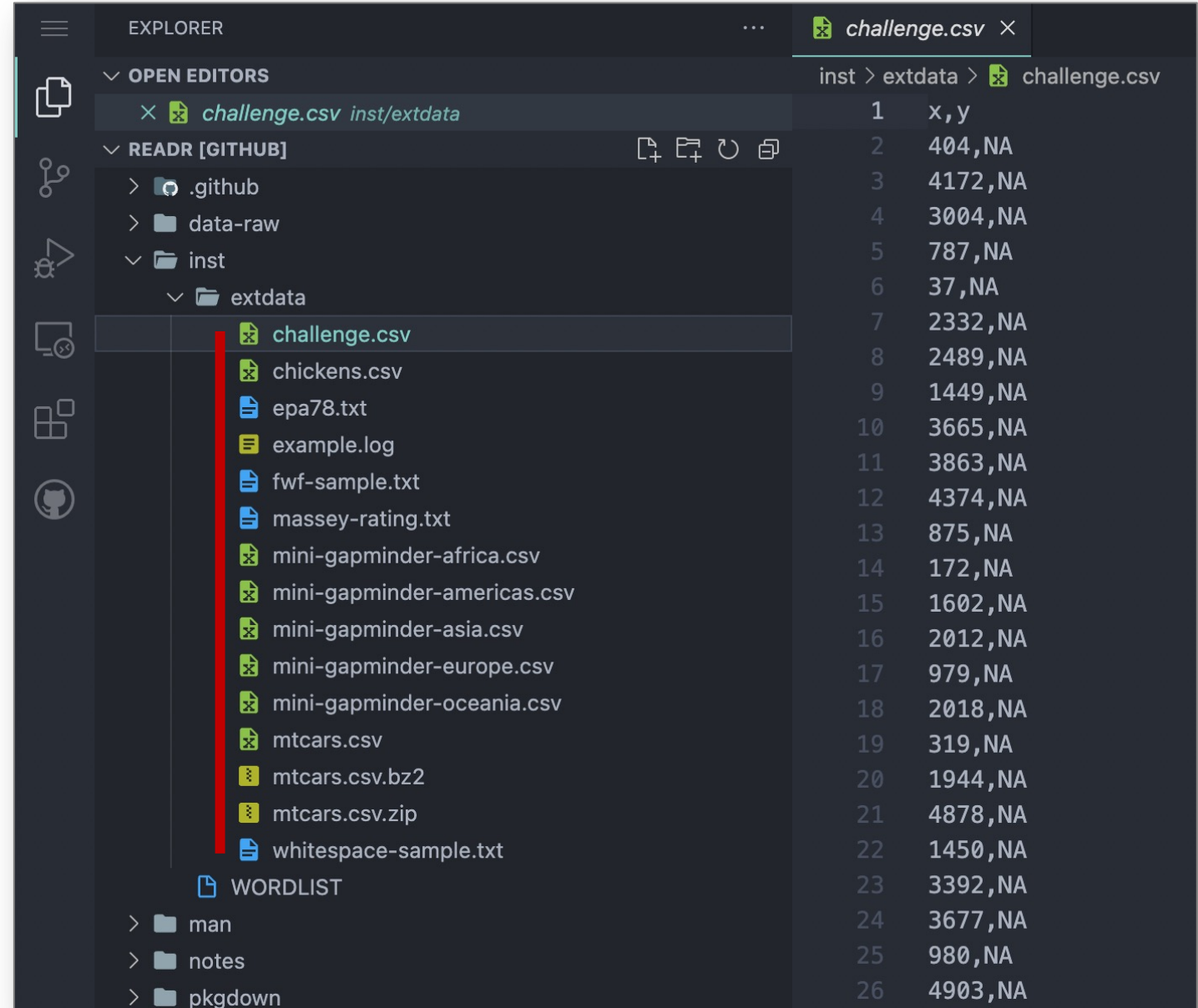
```
myPackage/  
  R/  
    functions.R  
    utils.R  
  man/  
    myfunction.Rd  
  tests/  
    testthat.R  
    testthat/  
      test-mvfun.R  
  inst/  
    extdata/  
      <raw-data-file>  
  DESCRIPTION  
  NAMESPACE  
  README.md  
  NEWS  
  LICENSE
```

Raw data

Raw data can be virtually any file, but it has to be relevant for the package development.

The main reason to include such files is when a key part of a package's functionality is to act on an external file (e.g. ``readr``, ``vroom``, ...).

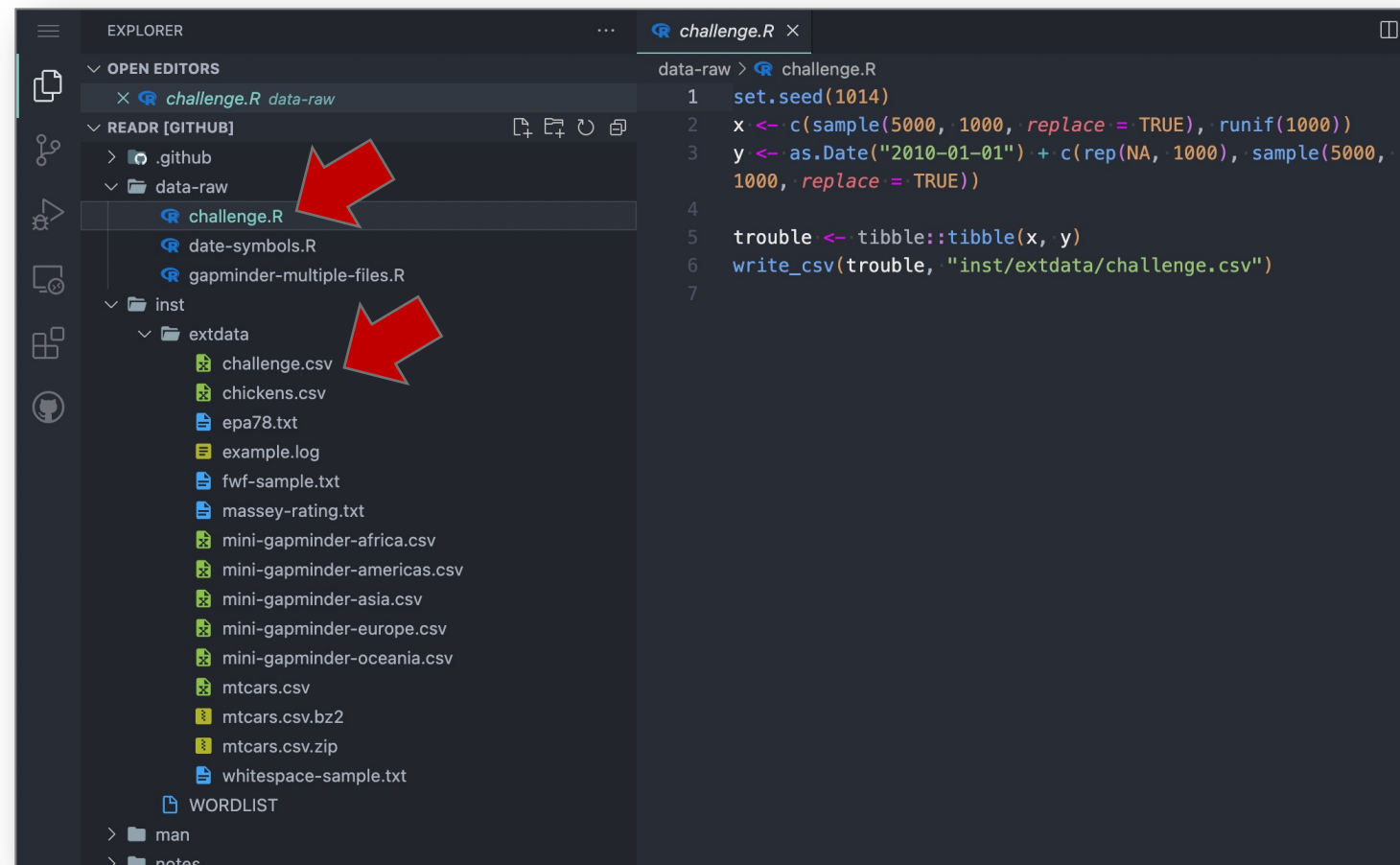
Raw data is stored in ``inst/extdata``.



Raw data

It is good practice to document how raw data files were generated.

You can do it by adding a `data-raw` folder to the root of your package and add R files describing how the data was created.



Raw data

It is good practice to document how raw data files were generated.

You can do it by adding a ``data-raw`` folder to the root of your package and add R files describing how the data was created.

Of note, Bioconductor specifically prefers these files to be located in ``inst/scripts/``:

<https://contributions.bioconductor.org/docs.html#doc-inst-script>.

Use raw data

Raw data should generally not be read by the end-user.

Instead, it is a way for the package writer to describe how to import files.

Use raw data

Raw data should generally not be read by the end-user.

Instead, it is a way for the package writer to describe how to import files.

For this reason, raw data is generally only a toy dataset, a small subset of an actual dataset (e.g. only a single chromosome out of a whole genome, etc.)

Use raw data

Raw data should generally not be read by the end-user.

Instead, it is a way for the package writer to describe how to import files.

For this reason, raw data is generally only a toy dataset, a small subset of an actual dataset (e.g. only a single chromosome out of a whole genome, etc.).

For this reason as well, if your package deals with already existing file formats (e.g. bed, bam, bigwig, ...), many BioC core packages (e.g. GenomicRanges, rtracklayer, Biostrings, ...) already provide toy datasets!!

Hosting all these raw data files has a cost (economical and environmental). Please, do check whether core packages can provide the type of files you'd need as a toy dataset.

Use raw data

The package writer can have access to their (or other packages') raw data files using `system.file()`:

```
## List raw data files shipped with GenomicRanges
> system.file('extdata', package = 'GenomicRanges') |> list.files()
[1] "feature_fragments.txt"

## Get full path to "feature_fragments.txt"
> system.file('extdata', "feature_fragments.txt", package = 'GenomicRanges')
[1] "/Users/jacques/Library/R/arm64/4.3/library/GenomicRanges/extdata/feature_fragments.txt"
```


Use raw data

The package writer can have access to their (or other packages') raw data files using ``system.file()`:`

GRangesList-class: GRangesList objects

In **Bioconductor/GenomicRanges**: Representation and manipulation of genomic intervals

Examples

```
## Construction with GRangesList():
gr1 <- GRanges("chr2", IRanges(3, 6),
               strand="+", score=5L, GC=0.45)
gr2 <- GRanges(c("chr1", "chr1"), IRanges(c(7,13), width=3),
               strand=c("+", "-"), score=c(3:4, GC=c(0.3, 0.5)))
gr3 <- GRanges(c("chr1", "chr2"), IRanges(c(1, 4), c(3, 9)),
               strand=c("-", "-"), score=c(6L, 2L), GC=c(0.4, 0.1))
grl <- GRangesList(gr1=gr1, gr2=gr2, gr3=gr3)
grl

## Summarizing elements:
elementNROWS(grl)
table(seqnames(grl))

## Extracting subsets:
grl[seqnames(grl) == "chr1", ]
grl[seqnames(grl) == "chr1" & strand(grl) == "+", ]

## Renaming the underlying sequences:
seqlevels(grl)
seqlevels(grl) <- sub("chr", "Chrom", seqlevels(grl))
grl
```

[illegible]

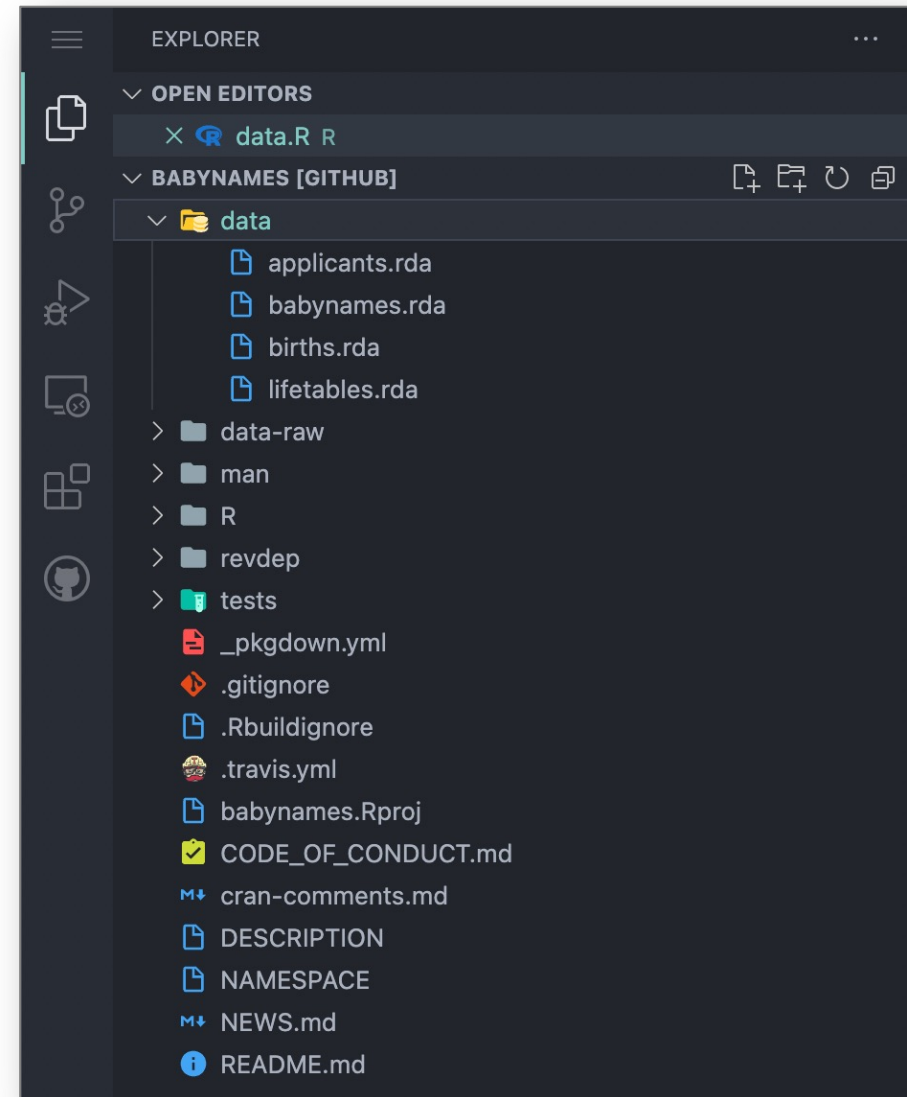
Processed data

Processed data are stored in the `data/` folder.

```
myPackage/  
  R/  
    functions.R  
    utils.R  
  man/  
    myfunction.Rd  
  tests/  
    testthat.R  
    testthat/  
      test-myfun.R  
  inst/  
    extdata/  
      <raw-data-file>  
  data/  
    mydata.rda  
  DESCRIPTION  
  NAMESPACE  
  README.md  
  NEWS  
  LICENSE
```

Processed data

Processed data are stored in the `data/` folder.



Processed data

Processed data are stored in the
`data/` folder.

The best way to provide processed
data in your package is through
`usethis::use\_data()`.

```
> chr <- vroom::vroom('HiCompute/testHiC.chr.tsv')
> chr
# A tibble: 17 x 4
  contig   length n_frags cumul_length
  <chr>    <dbl>   <dbl>      <dbl>
1 I       230218    1358         0
2 II      813184    4981       1358
3 III     316620    1948       6339
4 IV     1531933    9709       8287
5 V       576874    3484      17996
6 VI      270161    1734      21480
7 VII     1090940    6716      23214
8 VIII     562643    3405      29930
9 IX      439888    2756      33335
10 X       745751    4679      36091
11 XI      666816    4144      40770
12 XII     1078177    6728      44914
13 XIII     924431    5713      51642
14 XIV      784333    4847      57355
15 XV     1091291    6731      62202
16 XVI      948066    5814      68933
17 Mito     85779     160      74747
> usethis::use_data(chr)
v Saving 'chr' to 'data/chr.rda'
* Document your data (see 'https://r-pkgs.org/data.html')
```

Processed data

Processed data are stored in the
`data/` folder.

The best way to provide processed
data in your package is through
`usethis::use\_data()`.

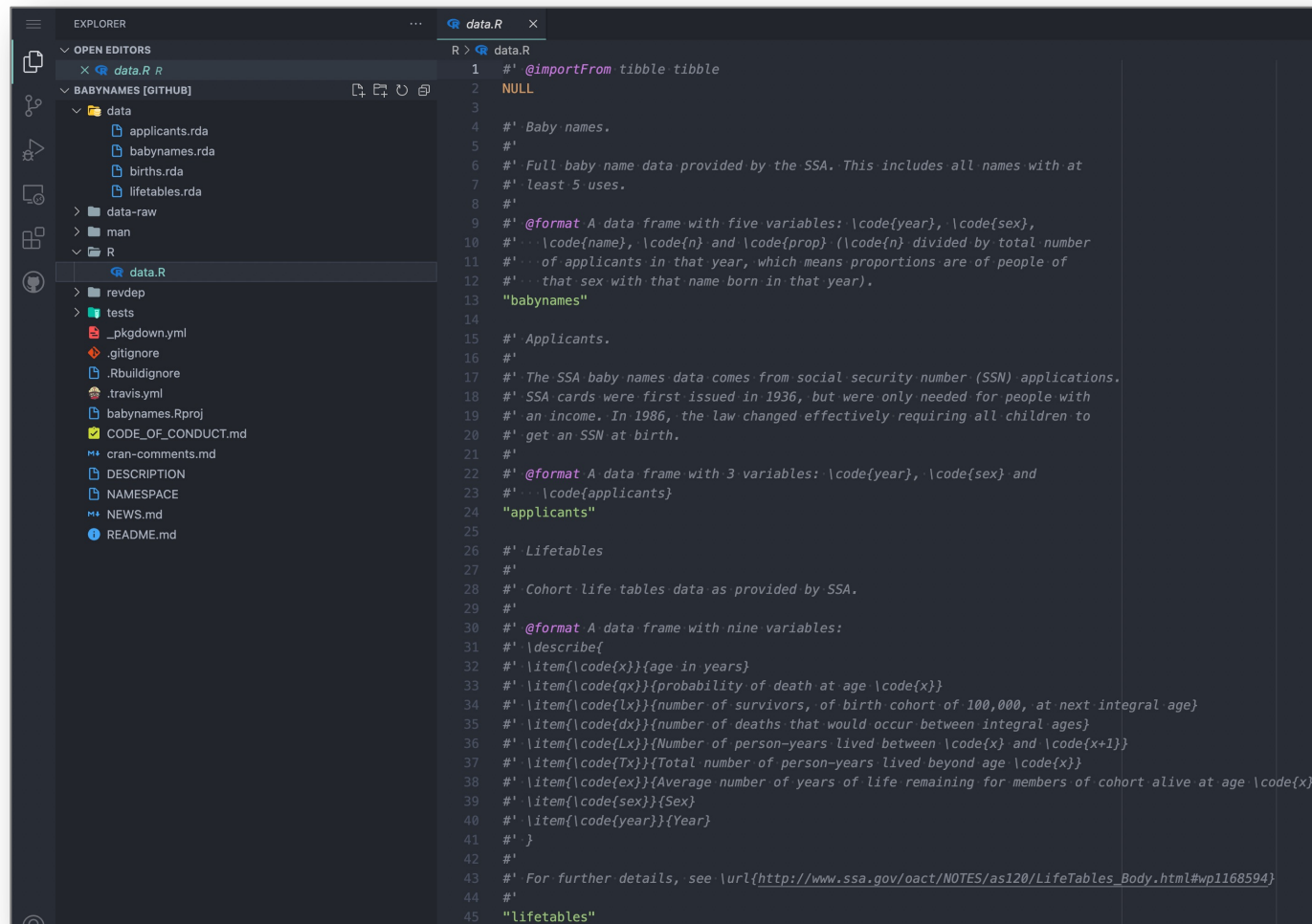
Avoid at all costs creating the `data/`
folder yourself. You should be able
to use `usethis::use\_data()` instead.

```
> chr <- vroom::vroom('HiCompute/testHiC.chr.tsv')
> chr
# A tibble: 17 x 4
  contig    length n_frags cumul_length
  <chr>    <dbl>   <dbl>      <dbl>
1 I        230218    1358         0
2 II       813184    4981       1358
3 III      316620    1948       6339
4 IV      1531933    9709       8287
5 V        576874    3484      17996
6 VI       270161    1734      21480
7 VII     1090940    6716      23214
8 VIII     562643    3405      29930
9 IX       439888    2756      33335
10 X       745751    4679      36091
11 XI      666816    4144      40770
12 XII     1078177    6728      44914
13 XIII    924431    5713      51642
14 XIV     784333    4847      57355
15 XV     1091291    6731      62202
16 XVI     948066    5814      68933
17 Mito     85779     160      74747
> usethis::use_data(chr)
v Saving 'chr' to 'data/chr.rda'
* Document your data (see 'https://r-pkgs.org/data.html')
```

Processed data

Like anything else in your package, your data should be documented.

The recommended way is to do that in a `R/data.R` file



```
1 #' @importFrom tibble tibble
2 NULL
3
4 #' Baby names.
5 #'
6 #' Full baby name data provided by the SSA. This includes all names with at
7 #' least 5 uses.
8 #'
9 #' @format A data frame with five variables: \code{year}, \code{sex},
10 #' \code{name}, \code{n} and \code{prop} (\code{n} divided by total number
11 #' of applicants in that year, which means proportions are of people of
12 #' that sex with that name born in that year).
13 "babynames"
14
15 #' Applicants.
16 #'
17 #' The SSA baby names data comes from social security number (SSN) applications.
18 #' SSA cards were first issued in 1936, but were only needed for people with
19 #' an income. In 1986, the law changed effectively requiring all children to
20 #' get an SSN at birth.
21 #'
22 #' @format A data frame with 3 variables: \code{year}, \code{sex} and
23 #' \code{applicants}
24 "applicants"
25
26 #' Lifetables
27 #'
28 #' Cohort life tables data as provided by SSA.
29 #'
30 #' @format A data frame with nine variables:
31 #' \describe{
32 #'   \item{\code{x}}{age in years}
33 #'   \item{\code{qx}}{probability of death at age \code{x}}
34 #'   \item{\code{lx}}{number of survivors, of birth cohort of 100,000, at next integral age}
35 #'   \item{\code{dx}}{number of deaths that would occur between integral ages}
36 #'   \item{\code{Lx}}{Number of person-years lived between \code{x} and \code{x+1}}
37 #'   \item{\code{Tx}}{Total number of person-years lived beyond age \code{x}}
38 #'   \item{\code{ex}}{Average number of years of life remaining for members of cohort alive at age \code{x}}
39 #'   \item{\code{sex}}{Sex}
40 #'   \item{\code{year}}{Year}
41 #' }
42 #'
43 #' For further details, see \url{http://www.ssa.gov/oact/NOTES/as120/LifeTables_Body.html#wp1168594}
44 #'
45 "lifetables"
```

Processed data

Like anything else in your package, your data should be documented.

The recommended way is to do that in a `R/data.R` file



```
R> data.R
1 #' @importFrom tibble tibble
2 NULL
3
4 #' Baby names.
5 #'
6 #' Full baby name data provided by the SSA. This includes all names with at
7 #' least 5 uses.
8 #'
9 #' @format A data frame with five variables: \code{year}, \code{sex},
10
11 #' Lifetables
12 #'
13 #' Cohort life tables data as provided by SSA.
14 #'
15 #' @format A data frame with nine variables:
16 #' \describe{
17 #' \item{\code{x}}{age in years}
18 #' \item{\code{qx}}{probability of death at age \code{x}}
19 #' \item{\code{lx}}{number of survivors, of birth cohort of 100,000, at next integral age}
20 #' \item{\code{dx}}{number of deaths that would occur between integral ages}
21 #' \item{\code{Lx}}{Number of person-years lived between \code{x} and \code{x+1}}
22 #' \item{\code{Tx}}{Total number of person-years lived beyond age \code{x}}
23 #' \item{\code{ex}}{Average number of years of life remaining for members of cohort alive at age \code{x}}
24 #' \item{\code{sex}}{Sex}
25 #' \item{\code{year}}{Year}
26 #' }
27 #'
28 #' For further details, see \url{http://www.ssa.gov/oact/NOTES/as120/LifeTables_Body.html#wp1168594}
29 #'
30 "lifetables"
```

Processed data

Processed data are made readily available to your package end-users through the ``data()`` function.

```
> library(babynames)
> data(babynames)
> babynames
# A tibble: 1,924,665 × 5
   year sex   name      n  prop
  <dbl> <chr> <chr>   <int> <dbl>
1  1880 F     Mary    7065 0.0724
2  1880 F     Anna    2604 0.0267
3  1880 F     Emma    2003 0.0205
4  1880 F   Elizabeth  1939 0.0199
5  1880 F    Minnie   1746 0.0179
6  1880 F   Margaret  1578 0.0162
7  1880 F      Ida    1472 0.0151
8  1880 F     Alice   1414 0.0145
9  1880 F    Bertha   1320 0.0135
10 1880 F     Sarah   1288 0.0132
# ... with 1,924,655 more rows
# i Use `print(n = ...)` to see more rows
```


Processed data

Processed data are made readily available to your package end-users through the `data()` function.

```
> data(lifetables)
> lifetables
# A tibble: 2,880 × 9
      x      qx      lx      dx      Lx      Tx      ex sex  year
  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <fct> <dbl>
1     0 0.146 100000 14596 90026 5151511 51.5 M    1900
2     1 0.0328 85404 2803 84003 5061484 59.3 M    1900
3     2 0.0163 82601 1350 81926 4977482 60.3 M    1900
4     3 0.0105 81251 855 80824 4895556 60.2 M    1900
5     4 0.00875 80397 703 80045 4814732 59.9 M    1900
6     5 0.00628 79693 501 79443 4734687 59.4 M    1900
7     6 0.00462 79193 366 79010 4655244 58.8 M    1900
8     7 0.00326 78827 257 78698 4576234 58.0 M    1900
9     8 0.00256 78569 201 78469 4497536 57.2 M    1900
10    9 0.00203 78368 159 78288 4419068 56.4 M    1900
# ... with 2,870 more rows
# i Use `print(n = ...)` to see more rows
```

Vignettes

Vignettes are extensive (comprehensive) walkthrough of your package functionalities.

Vignettes

Vignettes are extensive (comprehensive) walkthrough of your package functionalities.

They live in the `vignettes/` folder (duh 🤔) .

```
myPackage/  
  R/  
    functions.R  
    utils.R  
  man/  
    myfunction.Rd  
  tests/  
    testthat.R  
    testthat/  
      test-myfun.R  
  inst/  
    extdata/  
      <raw-data-file>  
  data/  
    <data>.Rda  
  vignettes/  
    myPackage.Rmd  
  DESCRIPTION  
  NAMESPACE  
  README.md  
  NEWS  
  LICENSE
```

Vignettes

Vignettes are extensive (comprehensive) walkthrough of your package functionalities.

They live in the ``vignettes/`` folder (duh 🙄) .

You can create a vignette boilerplate with either:

- `usethis::use_vignette('<YOUR-PACKAGE>')`
- `biocthis::use_bioc_vignette('<YOUR-PACKAGE>')`

Vignettes

Vignettes are extensive (comprehensive) walkthrough of your package functionalities.

They live in the ``vignettes/`` folder (duh 🙄) .

You can create a vignette boilerplate with either:

- `usethis::use_vignette('<YOUR-PACKAGE>')`
- `biocthis::use_bioc_vignette('<YOUR-PACKAGE>')`

You are not limited to a single vignette. However, you must provide at least one, which should be named `<YOUR-PACKAGE>.Rmd`.

Vignettes

Vignettes are extensive (comprehensive) walkthrough of your package functionalities.

They live in the `vignettes/` folder (duh 🙄) .

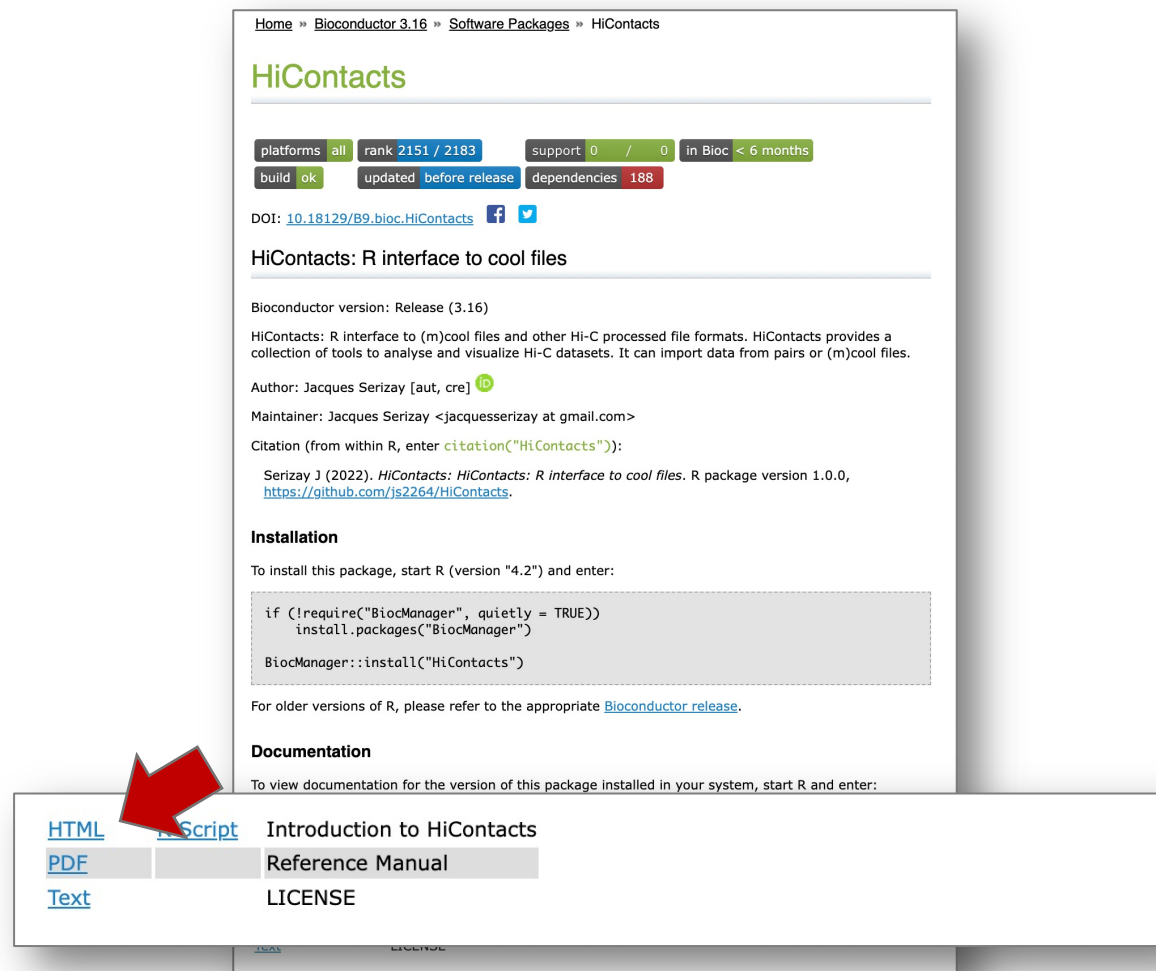
You can create a vignette boilerplate with either:

- `usethis::use_vignette('<YOUR-PACKAGE>')`
- `biocthis::use_bioc_vignette('<YOUR-PACKAGE>')`

You are not limited to a single vignette. However, you must provide at least one, which should be named `<YOUR-PACKAGE>.Rmd`.

Be as thorough as possible to describe all your package functionalities.

Once your package is accepted by BioC, your vignette will be compiled by the Bioconductor Single Package Builder into an HTML page, accessible through your package webpage.



The screenshot shows the Bioconductor package page for HiContacts. At the top, the breadcrumb navigation reads: Home » Bioconductor 3.16 » Software Packages » HiContacts. The package name "HiContacts" is displayed in green. Below it, a row of status badges includes: platforms all, rank 2151 / 2183, support 0 / 0, in Bioc < 6 months, build ok, updated before release, and dependencies 188. A DOI link (10.18129/B9.bioc.HiContacts) and social media icons for Facebook and Twitter are also present. The title "HiContacts: R interface to cool files" is followed by the Bioconductor version (Release (3.16)). The description states: "HiContacts: R interface to (m)cool files and other Hi-C processed file formats. HiContacts provides a collection of tools to analyse and visualize Hi-C datasets. It can import data from pairs or (m)cool files." The author is Jacques Serizay [aut, cre] and the maintainer is Jacques Serizay <jacquesserizay at gmail.com>. The citation is: Serizay J (2022). *HiContacts: R interface to cool files*. R package version 1.0.0, <https://github.com/js2264/HiContacts>. The "Installation" section provides instructions to start R (version "4.2") and enter the following code:

```
if (!require("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

BiocManager::install("HiContacts")
```

 For older versions of R, it refers to the appropriate Bioconductor release. The "Documentation" section instructs to start R and enter:

```
library(HiContacts)
```

 A red arrow points from the "HTML" link in the bottom left to the "Introduction to HiContacts" link in the bottom right. The bottom left also contains links for PDF and Text. The bottom right contains links for Introduction to HiContacts, Reference Manual, and LICENSE.

Home » Bioconductor 3.16 » Software Packages » HiContacts

HiContacts

platforms all rank 2151 / 2183 support 0 / 0 in Bioc < 6 months
build ok updated before release dependencies 188

DOI: [10.18129/B9.bioc.HiContacts](https://doi.org/10.18129/B9.bioc.HiContacts) [f](#) [t](#)

HiContacts: R interface to cool files

Bioconductor version: Release (3.16)

HiContacts: R interface to (m)cool files and other Hi-C processed file formats. HiContacts provides a collection of tools to analyse and visualize Hi-C datasets. It can import data from pairs or (m)cool files.

Author: Jacques Serizay [aut, cre] [iD](#)

Maintainer: Jacques Serizay <jacquesserizay at gmail.com>

Citation (from within R, enter `citation("HiContacts")`):

Serizay J (2022). *HiContacts: R interface to cool files*. R package version 1.0.0, <https://github.com/js2264/HiContacts>.

Installation

To install this package, start R (version "4.2") and enter:

```
if (!require("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

BiocManager::install("HiContacts")
```

For older versions of R, please refer to the appropriate [Bioconductor release](#).

Documentation

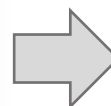
To view documentation for the version of this package installed in your system, start R and enter:

```
library(HiContacts)
```

[HTML](#) [R Script](#) [PDF](#) [Text](#)

[Introduction to HiContacts](#) [Reference Manual](#) [LICENSE](#)

```
HiContacts.Rmd x
vignettes > HiContacts.Rmd > abc unnamed-chunk-1
1 ---
2 title: "Introduction to HiContacts"
3 author: "Jacques Serizay"
4 date: "`r Sys.Date()`"
5 output:
6   BiocStyle::html_document
7 vignette: >
8   %\VignetteIndexEntry{Introduction to HiContacts}
9   %\VignetteEngine{knitr::rmarkdown}
10  %\VignetteEncoding{UTF-8}
11 ---
12
13 Select Chunk | Run Chunk
14 {r, eval = TRUE, echo=FALSE, results="hide", warning=FALSE}
15 knitr::opts_chunk$set(
16   collapse = TRUE,
17   comment = "#>",
18   crop = NULL
19 )
20 suppressPackageStartupMessages({
21   library(ggplot2)
22   library(dplyr)
23   library(GenomicRanges)
24   library(HiContactsData)
25   library(HiContacts)
26 })
27 ```
```



Introduction to HiContacts

Jacques Serizay

2022-11-01

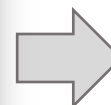
Contents

- 1 Getting started
 - 1.1 The `Contacts` class
 - 1.2 Basics: importing `.(m)/cool` files as `Contacts` objects
 - 1.3 Slots
 - 1.4 Slot setters
 - 1.4.1 Scores
 - 1.4.2 Features
 - 1.5 Coercing `Contacts`
- 2 Plotting matrices
 - 2.1 Plot matrix heatmaps
 - 2.2 Plot loops
 - 2.3 Plot borders
- 3 Arithmetics
 - 3.1 Subsetting a contact map
 - 3.2 Computing autocorrelated contact map
 - 3.3 Detrending contact map (map over expected)
 - 3.4 Summing two maps
 - 3.5 Computing ratio between two maps
- 4 Contact map analysis
 - 4.1 Virtual 4C
 - 4.2 Cis-trans ratios
 - 4.3 P(s)
- 5 Session info

Vignettes

```
HiContacts.Rmd ×
vignettes > HiContacts.Rmd > abc unnamed-chunk-1

28 ∨ # Getting started
29
30 ∨ ## The `Contacts` class
31
32 `HiContacts` package implements the new `Contacts` S4 class. It is build
33 on pre-existing Bioconductor classes, namely `InteractionSet`,
34 `GenomicInteractions` and `ContactMatrix`
35 (`Lun, Perry & Ing-Simmons, F1000Research 2016`), and leverages them to
36 import locally stored `(m)cool` files. It further provides **analytical**
37 and **visualization** tools to investigate contact maps directly in `R`.
38
39 Select Chunk | Run Chunk
39 ∨ ```{r}
40 showClass("Contacts")
41 contacts <- contacts_yeast()
42 contacts
43 ```
44
45 Select Chunk | Run Chunk
45 ∨ ```{r}
46 citation("HiContacts")
47 ```
```



1 Getting started

1.1 The `Contacts` class

`HiContacts` package implements the new `Contacts` S4 class. It is build on pre-existing Bioconductor classes, namely `InteractionSet`, `GenomicInteractions` and `ContactMatrix` (Lun, Perry & Ing-Simmons, F1000Research 2016), and leverages them to import locally stored `(m)cool` files. It further provides **analytical** and **visualization** tools to investigate contact maps directly in `R`.

```
showClass("Contacts")
#> Class "Contacts" [package "HiContacts"]
#>
#> Slots:
#>
#> Name:          fileName          focus          resolutions
#> Class:          character        characterOrNULL numeric
#>
#> Name:          resolution        interactions    scores
#> Class:          numeric          GInteractions SimpleList
#>
#> Name: topologicalFeatures        pairsFile        metadata
#> Class: SimpleList characterOrNULL list
#>
#> Extends: "Annotated"
contacts <- contacts_yeast()
#> snapshotDate(): 2022-10-24
#> see ?HiContactsData and browseVignettes('HiContactsData') for documentation
#> loading from cache
contacts
#> `Contacts` object with 74,360 interactions over 802 regions
#> -----
#> fileName: "/home/biocbuild/.cache/R/ExperimentHub/37cabfdcee0b5_7752"
#> focus: "II"
#> resolutions(5): 1000 2000 4000 8000 16000
#> current resolution: 1000
#> interactions: 74360
#> scores(2): raw balanced
#> topologicalFeatures: loops(0) borders(0) compartments(0) viewpoints(0)
#> pairsFile: N/A
#> metadata(0):
```

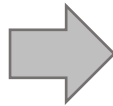
```
citation("HiContacts")
#>
#> To cite package 'HiContacts' in publications use:
#>
#> Serizay J (2022). _HiContacts: HiContacts: R interface to cool
#> files. R package version 1.0.0,
#> <https://github.com/js2264/HiContacts>.
#>
#> A BibTeX entry for LaTeX users is
#>
#> @Manual{,
#>   title = {HiContacts: HiContacts: R interface to cool files},
#>   author = {Jacques Serizay},
#>   year = {2022},
#>   note = {R package version 1.0.0},
#>   url = {https://github.com/js2264/HiContacts},
#> }
```

Vignettes

This means all the code in a vignette must work!

Standard package content

```
myPackage/  
  R/  
    functions.R  
    utils.R  
  man/  
    myfunction.Rd  
  tests/  
    testthat.R  
    testthat/  
      test-myfun.R  
DESCRIPTION  
README.md  
NAMESPACE  
NEWS  
LICENSE
```



```
myPackage/  
  R/  
    functions.R  
    utils.R  
  man/  
    myfunction.Rd  
  tests/  
    testthat.R  
    testthat/  
      test-myfun.R  
  inst/  
    extdata/  
      <raw-data-file>  
  data/  
    <data>.Rda  
  vignettes/  
    myPackage.Rmd  
DESCRIPTION  
NAMESPACE  
README.md  
NEWS  
LICENSE
```